HR ANALYTICS

Employee Attrition & Workforce Intelligence

Business Problem Statement

End-to-End Data Analytics Portfolio Project

Field	Details
Dataset	HR Analytics Employee Attrition Dataset
Total Records	1,470 Employees
Total Features	27 Variables
Tools Used	Python, MySQL, Power BI
Project Type	End-to-End Analytics Portfolio Project
Target Variable	Attrition (Employee Left: Yes / No)
Domain	Human Resources Analytics

1. Background

Employee attrition — the rate at which employees leave an organisation — is one of the most critical and costly challenges facing modern businesses. When a skilled employee leaves, the organisation bears not only the direct cost of recruiting and training a replacement, but also the hidden costs of lost institutional knowledge, reduced team productivity, and declining morale among remaining staff.

According to industry benchmarks, replacing a single employee can cost anywhere between 50% to 200% of their annual salary depending on their seniority and role. For large organisations with hundreds of employees, even a modest attrition rate of 15-20% can translate into millions of dollars in annual losses.

Despite the scale of this problem, many HR departments still rely on gut feeling, exit interviews, or lagging indicators to understand why employees leave. This project takes a data-driven approach to answer the same questions more accurately, earlier, and at scale.

2. Problem Statement

The organisation represented in this dataset is experiencing a 16% employee attrition rate — meaning roughly 1 in every 6 employees leaves the company. This figure is above the industry average and is causing measurable disruption across departments, particularly in Sales and Human Resources.

The core business questions this project aims to answer are:

- Who is leaving — which departments, job roles, age groups, and demographic segments have the highest attrition?
- Why are they leaving — is it driven by low salary, poor work-life balance, lack of promotion, overtime pressure, or low job satisfaction?
- Is the pattern statistically significant — or could the observed differences be due to random variation in a small sample?
- Which employee groups are at the highest risk of leaving in the near future?
- What concrete actions can HR and management take to reduce attrition based on the data?

3. Business Objectives

This project is structured around three primary business objectives:

3.1 Understand Attrition Patterns

Identify which segments of the workforce are most affected by attrition and quantify the attrition rate across every relevant dimension — department, job role, age group, marital status, overtime status, and business travel frequency.

3.2 Identify Root Causes

Determine which variables — salary, satisfaction scores, years since last promotion, overtime, work-life balance, performance rating, and tenure — are most strongly associated with employees leaving. Validate findings statistically using hypothesis testing to ensure conclusions are reliable and not coincidental.

3.3 Deliver Actionable Recommendations

Translate data findings into specific, prioritised recommendations that an HR manager or business leader can act on immediately — including compensation reviews, promotion policy changes, overtime management, and targeted retention programmes for high-risk groups.

4. Dataset Overview

The dataset used in this project is the HR Analytics Employee Attrition dataset, a widely used benchmark dataset in the HR analytics domain. It contains records for 1,470 employees across three departments and nine job roles, with 27 variables covering demographics, compensation, satisfaction, and career history.

Category	Variables Included
Demographics	Age, Gender, MaritalStatus, EducationField
Compensation	MonthlyIncome, DailyRate, HourlyRate, MonthlyRate, PercentSalaryHike
Job Information	Department, JobRole, JobLevel, BusinessTravel, OverTime, StandardHours
Satisfaction Scores	JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance, RelationshipSatisfaction
Career History	TotalWorkingYears, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, NumCompaniesWorked
Performance	PerformanceRating, TrainingTimesLastYear, JobInvolvement
Target Variable	Attrition (Yes = Left, No = Stayed)

5. Analytical Approach

This project follows a structured end-to-end analytics pipeline across three tools, with each tool handling the tasks it is best suited for:

Phase	Tool	Tasks Performed
Data Cleaning	Python (Pandas)	Null checks, duplicate removal.
Business Analysis	MySQL	Aggregation queries, attrition rates by segment, salary summaries, promotion gap analysis, satisfaction breakdowns
Statistical Testing	Python (SciPy)	2-sample t-test on salary vs attrition, one-way ANOVA on salary across departments, correlation heatmaps
Visualisation (EDA)	Python (Matplotlib, Seaborn)	Boxplots, distribution plots, heatmaps, countplots, group comparison charts
Dashboard	Power BI	5-page interactive report with KPI cards, slicers, and advanced visuals including treemap, waterfall, funnel, scatter, and ribbon charts

6. Key Business Questions Addressed

Attrition Analysis

- What is the overall attrition rate and how does it compare across departments?
- Which job roles have the highest number of employees leaving?
- Do younger employees (under 30) leave at a significantly higher rate than older groups?
- Does working overtime significantly increase the likelihood of an employee leaving?
- Which marital status group shows the highest attrition rate?

Salary & Compensation

- Is there a statistically significant salary difference between employees who left and those who stayed?
- Which job roles and departments have the highest and lowest average salaries?
- Are high-performing employees receiving proportionally higher salary hikes?
- How is monthly income distributed across the organisation?

Satisfaction & Engagement

- Do employees with low job satisfaction leave more frequently?
- Which department has the highest average job satisfaction score?
- Does working overtime negatively impact work-life balance and job satisfaction scores?
- Is there a relationship between environment satisfaction and attrition?

Career Growth & Retention Risk

- Which department has the longest average wait time between promotions?
- Do employees who have worked at more companies previously tend to leave sooner?
- Is there a correlation between years at the company and monthly income?
- Which combination of factors identifies the highest-risk flight-risk employee group?